THE EMS MAPPING PROJECT

User Guidebook

A step-by-step reference for setting up and navigating your custom EMS map!

The University of Texas at Dallas

University Emergency Medical Response (UEMR)

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Hello!

Welcome to the EMS Mapping Project, a free website that can create custom heatmaps and evaluate a virtually unlimited range of parameters pertaining to your agency's operations! Roshan, Divya, and I sincerely hope you are able to derive as much use from this site as our agency, UT Dallas University Emergency Medical Response, has! (Go Comets! Whoosh! 📎)

This comprehensive guide details exactly how to make use of The EMS Mapping Project. Each section below details a different feature of the program. (The tabs/navigation guide on the left side may be of use!).

I would recommend reading through this guide at least once before using the program for yourself or your organization to gain some familiarity with how everything works. If you would like to play with all of the features of the program before uploading your incident data, I have included a sample data set with some random parameters and data attached to this document that you may download to follow along with the guide as I go through and explain each feature!

If any unanticipated issues arise with the website, or if you have any questions about the program that are not answered in this guide, please feel free to let me know at the following email address: priyadarbha.bchm@gmail.com.

Best regards,

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1. Overview & Interface

The EMS Mapping Project is a browser-based, geographic analysis tool for Emergency Medical Services agencies. It lets you overlay incident data onto a live map, delineate custom coverage regions, filter data by any parameter, and visualize crews' response-time efficiency.

Interface at a Glance

Feature / Button	What it does
Left Sidebar	All the controls you'll need are here. The two tabs, ⚙ Settings and 📊 Analysis, separate tge setup from the analytics.
Map Canvas (remainder of screen)	An OpenStreetMap view for drawing areas and viewing color-coded heatmaps.
Settings Tab	Map Center · Station & Routing · Incident Data upload · Area Selection · Categories · Import/Export · Reset All.
Analysis Tab	View Options · Heatmap Mode · Legend · Data Filters · Custom Filters · Summary Panel.

💡 All your work (areas, CSV incident data, station location, filters) is automatically saved to your browser's local storage. Refreshing the page will not lose your data.
Use Export All Settings (Section 10) to back up or share your configuration.

2. Setting a Map Center

Before drawing any areas, position the map over your operational zone. The default view opens centered on The University of Texas at Dallas. All three Map Center controls are under the Map Center in the Settings tab.

Available Controls

Feature / Button	What it does
Set Map Center	Prompts for an address, landmark, city, or ZIP code to recenter your map view.
Set Current View as Default	Saves the exact position and zoom you are currently viewing. Use this after moving and zooming to exactly the right vantage point you desire.
Go to Default	Instantly returns the map to your saved default without altering any drawn areas.

Step-by-Step: Changing the Map Center

Step 1: In the Settings tab, expand Map Center.

Step 2: Click Set Map Center and type a search term (e.g., "800 W Campbell Rd" or "75080").

Step 3: A confirmation dialog shows theaddress and coordinates. Click OK to accept the move.

Step 4: Adjust zoom with the scroll wheel if needed, then click Set Current View as Default to lock it in.

> If a location cannot be found, add more information, like a city or state, to the search.
> The zoom level is not changed when you use Set Map Center.
> Go to Default is your 'home' button if you accidentally navigate too far away.

3. Station & Routing Configuration

The station marker defines the origin for all distance and travel-time calculations. You only need this section if you plan to use the Distance/Time Efficiency heatmap.

Placing the Station Marker

Step 1: Expand Station & Routing in the Settings tab.

Step 2: Click Set Station Location.

Step 3: Click any map point to drop the marker. A 'Station' popup opens automatically.

Step 4: Drag the marker anywhere to fine-tune its position. The coordinates update in real time.

Step 5: To delete the marker, right-click it.

Routing Modes

Feature / Button	What it does
Bike / Golf Cart / Walk (default)	Uses bicycle-network routing via the OSRM API. Best for campus environments where units travel exclusively on walkable paths/sidewalks and bike lanes.
Drive	Uses road-network routing. Appropriate when units respond by car on public streets.

Switching routing modes after distances are calculated triggers a fresh OSRM calculation next time Distance/Time mode is activated.
The station location and routing preference are both saved across browser refreshes.

4. Uploading Incident Data (CSV)

Incident data powers every analysis feature. Data should be uploaded as a standard comma-separated values (.csv) file. Include as many columns as you like; the application automatically classifies each as categorical or numerical.

How to Upload

Step 1: In the Settings tab, expand Incident Data.

Step 2: Click the file input and select your .csv file.

Step 3: An alert will show the number of rows imported, categorical filters detected, and the name of the location column.

Step 4: The Data Filters section populates automatically with collapsible checkbox groups for each categorical column.

Required Column: location

The program matches each incident row to a drawn area by comparing the value in the location column to the name you gave that area. Column name detection is case-insensitive, but the location IDs and areas ARE case-sensitive for matching purposes.

Optional Column: timeElapsed

Including a numeric 'timeElapsed' column (in minutes) unlocks the Distance/Time Efficiency heatmap. The values in this column are averaged per area.

Column Type Auto-Detection

Feature / Button	What it does
Text / categorical columns	Become collapsible checkbox filter groups in the Data Filters panel.
Numeric columns	Skipped by standard filters. Create a Custom Filter for these using numeric ranges.
Location column	Used for area-matching only; not shown as a filter.
timeElapsed column	Averaged per area; required for Distance/Time heatmap.

CSV data survives page refreshes via local storage.
To remove CSV data without deleting drawn areas, click Clear Incident Data Only.
Column headers are whitespace-trimmed — 'Day ' and 'Day' are treated identically.
Avoid commas inside unquoted cell values; quote the cell if needed.
Numerical filters may require a custom filter to establish desired numerical ranges.

5. Area Selection & Polygon Drawing

Areas are the primary unit of analysis. Each area's name must match a value in the location column of your CSV. The program counts how many rows share the area name and colors it on the heatmap accordingly.

Drawing a New Area

Step 1: Expand Area Selection in the Settings tab.

Step 2: Click Start New Area.

Step 3: Click on the map to place the first vertex. A red circle appears.

Step 4: Continue clicking to add vertices. A dashed red outline previews the shape.

Step 5: Hold and drag any vertex to reposition it — the polygon outline updates live.

Step 6: Click Finish, then enter a name that exactly matches the location column value.

Names are case-sensitive: 'Lab1' and 'lab1' will NOT match.

Editing, Deleting & Clearing

Feature / Button	What it does
Drag a vertex	Click-hold any red circle, drag to new position, release.
Ctrl+Z / Cmd+Z (while drawing)	Removes the most recently placed vertex.
Escape (while drawing)	Cancels the entire drawing session, removing all placed points and the preview polygon.
Right-click a vertex (while drawing)	Deletes that individual point (minimum 3 required).
Right-click inside a finished polygon	Deletes the entire area.
Clear All Areas	Removes all regular area at once. Categories do not clear.

Exporting & Importing Areas

Click Export Areas to download a .json file of all drawn areas (and categories). Upload it via the Upload Areas file input to restore drawings on any computer.

Area names must match CSV location values exactly (case-sensitive).
It's fine to have overlapping; each will be counted independently.
Labels float can be hidden in the Analysis tab.
Drawn areas do not delete after browser refreshes and are stored locally.

6. Category Creation

Categories let you combine data from multiple locations under a single region so you can avoid creating individual areas for each location ID. For example, if your CSV has location IDs for Lab1, Lab2, and Lab3 as separate areas, but you want to analyse all labs as one unit, called "Lab," draw a category polygon and assign all three location IDs to it.

Enabling Category Mode

Step 1: Expand 🗀 Categories in the Settings tab.

Step 2: Click Enable Category Mode to only show the categories on the map.

Drawing a Category

Step 1: Ensure a CSV file is loaded

Step 2: Click Start New Category.

Step 3: Use the search box to find location IDs, then check every location that belongs here.

Step 4: Click on the map to draw the boundary (same process as regular areas).

Step 5: Click Finish Category to save and name it.

Clearing & Deleting Categories

Feature / Button	What it does
Right-click inside a category polygon	Delete that single category.
Clear All Categories	Removes all categories (does not affect existing areas).

How Categories Aggregate Data

Feature / Button	What it does
Incident count	Total filtered rows across all assigned locations.
Average timeElapsed	Mean response time across all incidents in all assigned locations.
Distance / Efficiency ratio	Calculated from category centroid to the station.

Category labels appear in gold on the map. Label visibility is toggled in the Analysis tab.

🗀 Categories persist in local storage even when Category Mode is turned off.
When Category Mode is off, categories are hidden but NOT deleted.
A location ID can belong to multiple categories: counts are independent.
Escape cancels an in-progress category drawing the same as for regular areas.

7. Custom Filters

Custom Filters extend the standard checkbox filters to support value grouping. They are created in the Analysis tab's Custom Filters panel.

Opening the Custom Filter Builder

Step 1: Upload a CSV file first.

Step 2: Switch to the Analysis tab, then expand Custom Filters.

Step 3: Click + Create Custom Filter.

Step 1: Column & Name

Select the CSV column to filter on from the Column dropdown. Then type a display name for the filter (e.g., 'Age Groups' or 'Priority').

Step 2: Filter Type

Feature / Button	What it does
Numeric Ranges (default)	Define numeric value ranges using semicolon-separated expressions. Example: <2;2-4.5;>4.5
Categorical Groups	Check one or more values from the column, enter a group name, and click + Add Group. Repeat for each group you want to create. Example: check 'Blunt Trauma', 'Penetrating Trauma', and 'Hemorrhage' and name the group Trauma'.

Using a Custom Filter

Custom filters work identical to regular filters. Select the desired filter condition and watch your heatmap update in real time.

Feature / Button	What it does
Delete a Custom Filter	Click the Delete button on the filter card.
Clear all active ranges	Click Clear All Filters in the Analysis tab to reset everything simultaneously.

Custom Filters are saved to local storage and survive page refreshes.
Custom Filters AND with standard Data Filters — both must pass for a row to count.
For numeric filters: non-numeric cell values are excluded when any range is active.
For categorical filters: the comparison is exact and case-sensitive.

8. Analysis Tab: View Options & Heatmap Modes

Switch to the 📊 Analysis tab to control how data is visualized on the map.

View Options

Feature / Button	What it does
Show Area Labels	Toggles the name labels of each regular area.
Show Category Labels	Toggles the gold name labels for categories (only visible when categories exist).

Heatmap Mode: Incident Count (default)

Areas are shaded on a gray-to-blue gradient proportional to the number of filtered incidents matching each area. The highest-count area is deepest blue, and areas with zero incidents are gray.

Heatmap Mode: Distance / Time Efficiency

Areas are coloured on a red-to-green gradient where green indicates high miles-per-minute efficiency (fast relative response time) and red indicates lower efficiency. Requires both a station marker and a 'timeElapsed' column in the CSV.

Feature / Button	What it does
Efficiency Ratio formula	Distance from station to area centroid (miles) ÷ Average timeElapsed (minutes)

Distance Calculation & Caching

When you first switch to Distance/Time mode, a 'Calculating distances…' spinner appears. This may take a few minutes, depending on the number of areas and categories you have loaded. Maps with greater than 50 areas/categories frequently take up to two minutes to calculate. Because all calculated distances and results are stored locally, however, switching modes again is instant unless you move the station, switch routing profiles, or reset any part of the data.

Legend

The Legend panel always reflects the current heatmap mode. In Incident Count mode: gray-to-blue. In Distance/Time mode: red-to-green.

🗺 Switching heatmap mode while filters are active immediately recolors the map.
Click any area polygon to open a popup with full incident and filter breakdowns.
The popup shows filtered vs. total counts side-by-side when filters are active as 'filtered (total)'.

9. Data Filters & the Summary Panel

The Analysis tab's Data Filters section contains auto-generated checkbox groups for every categorical CSV column. The Summary panel below it provides a quick text overview of the currently loaded data.

Using Data Filters

Expand any column group and check one or more of the filters to restrict tabulated incidents to rows matching at least one checked value in that group. An incident must satisfy all active filter groups to be counted.

Feature / Button	What it does
Day-of-week columns (Su/M/T/W/Th/F/Sa)	Automatically sorted in calendar order rather than alphabetically.
Clear All Filters button	Unchecks every standard filter and every custom filter range at once.

Clicking an Area for Detail

Clicking any area or category polygon opens a popup containing:

- Area/Category name.
- Filtered incident count (with totals in parentheses when filters are active).
- Per-filter breakdowns showing value counts for all filterable conditions.
- A scrollable table of all filtered incidents for that area.
- An 'All Incidents (Unfiltered)' table that displays all incidents counted in that area/category
- Distance/Time metrics only when that heatmap mode is selected.

Summary Panel

The Summary box updates instantly on every filter change. In Incident Count mode it shows the top 3 areas by count. In Distance/Time mode it shows efficiency rankings, or just the top 3 fastest and bottom 3 slowest when more than 6 areas exist.

> Filter changes apply instantly.
> Popup filter comparisons are useful for shift-vs-shift or day-vs-day analysis of a particular area.
> In Distance/Time mode the popup shows both filtered and unfiltered efficiency.

10. Import and Export Settings

The program offers two levels of export. Use them to back up your work, share configurations with others, or transfer settings between computers.

Feature / Button	What it does
Export Areas (Settings → Area Selection)	Downloads a .json file of drawn areas and categories only. Re-import with the Upload Areas file input.
Export All Settings (Settings → Import/Export)	Downloads a .json file including areas, categories, custom filters, map center, station, and routing mode. Recommended for backing up data, custom filters, and drawn areas/categories.
Import All Settings (Settings → Import/Export)	Uploads an all-settings .json, replacing all current polygons, categories, filters, station, and map center. No incident .CSV data is stored in any export for institutional/organizational privacy control.
Clear Incident Data Only (Settings → Incident Data)	Wipes incident .CSV data rows while keeping all drawn areas, categories, and filters intact.
🪣 **Clear All Areas (Settings → Area Selection)**	Removes all regular areas while keeping categories.
🪣 **Clear All Categories (Settings → Categories)**	Removes all categories while keeping regular areas.
⚠ **Reset All (bottom of Settings tab)**	Removes everything: polygons, categories, incident .CSV rows, custom filters, distances. Map center default and station are preserved.

Local Storage Features

- Refreshing the page does not lose any data.
- Data is device- and browser-specific: it does not sync across devices.
- Clearing browser history or using private/incognito mode will erase local storage.
- Use the Export All Settings button as an off-browser backup.

> Always export your settings before clearing browser data or switching browsers.
> CSV row data is NOT included in any export to allow organizations to maintain control over the privacy of their data.

11. Keyboard Shortcuts & Quick Tips

Feature / Button	What it does
Ctrl+Z / Cmd+Z (while drawing)	Undo the last placed vertex. Works for areas and categories.
Esc (while drawing)	Cancel the entire drawing session instantly, removing all placed points.
Right-click on a vertex (while drawing)	Delete that individual vertex.
Right-click inside a finished polygon	Deletes the area or category.
Right-click the station marker	Deletes the station marker.
Scroll wheel on the map	Zoom in / out.
Click-drag the map background	Pan the map.
Click inside a finished polygon	Opens the incident detail popup for that area.

General Tips

- Name areas **exactly** as they appear in your CSV location column: spelling and capitalization matter.
- You may draw areas before or after uploading CSV data. The order does not matter.
- Use the Esc key to cleanly cancel a drawing session if you change your mind mid-way.
- Use categories when multiple location IDs represent a single logical zone.
- Clear All Areas / Clear All Categories are faster than right clicking each individually.
- After uploading a new CSV, the calculated distances are cleared automatically.
- Toggle between heatmap modes freely, the map recolors instantly.
- The guidebook link in the sidebar header opens this PDF for quick reference.

12. CSV Format Reference

Below is a minimal example CSV that exercises every application feature, so you can mess around with each feature on a small, random data set to get the hang of all the features of this program.

location	day	shift	chief complaint	age	timeElapsed (min)
Lab1	M	Morning	Chest Pain	22	4.2
Lab2	T	Evening	Syncope	45	6.1
Gym	F	Morning	Lacerations	19	3.5
Cafeteria	Sa	Night	Allergic Reaction	31	7.8

Column Rules Summary

Feature / Button	What it does
location (required)	Must match the area name assigned when drawing areas. Case-sensitive.
timeElapsed (optional)	By default, measured in minutes. Required for Distance/Time heatmap. Any non-numeric values are ignored.
Text-based columns (e.g., chief complaint)	Become custom filter groups in the Analysis tab.
Numeric columns (e.g., age)	Not auto-filtered. Create a Custom Filter using numeric ranges to generate desired numerical buckets.
Quoting	Quote data that contain commas (e.g., "Tightness, Chest")

13. Troubleshooting

Feature / Button	What it does
Heatmap shows all areas gray after uploading CSV	Area names must match the location column exactly (case-sensitive). Click an area to open its popup and confirm the match.
Distance/Time mode alert: 'CSV must include timeElapsed column'	Add a numeric column named 'timeElapsed' (exact spelling, case-insensitive) to your CSV.
Distance/Time mode alert: 'Please set a station location first'	Go to Settings → Station & Routing and place a station marker on the map.
'Calculating distances…' spinner runs for a long time	The OSRM is a free, public service. Each area requires a separate calculation request with a 100 ms delay. Be patient, especially with many areas.
Distances look wrong	Try switching routing modes (Bike/Walk vs. Drive). Also verify the station marker is at the correct position.
Custom Filter (Numeric) shows no results	Confirm the column actually contains numeric values. Non-numeric rows are excluded when any numeric range is active.
Custom Filter (Categorical) shows no results	The comparison is exact and case-sensitive. Check that the group values exactly match what appears in the CSV column.
Import All Settings fails: 'Incompatible settings file format'	You may be uploading an area-only export to the Import All Settings input. Use the matching import button for each file type.
Data disappears after clearing browser history	Local storage is cleared with browsing data. Export your settings beforehand to save them.
Map tiles missing / OSRM not responding	Both features require internet. Check your network connection.
Category popup shows 0 incidents despite data	Ensure location IDs in the category exactly match incident .CSV location values, and that no filters are active that are excluding all the rows.
Escape key not cancelling the drawing	Click somewhere on the map first to ensure the page has keyboard focus, then press Escape.

🛠 For persistent issues, try Reset All, then re-import your exported settings file to rule out corrupted local storage as the cause.

Quick-Reference Card

A fast reminder of the most common workflows.

Feature / Button	What it does
First-time setup	1. Set Map Center → 2. Set Station Location → 3/4. Upload CSV and Draw Areas/Categories → 5. Analyze
Match CSV to areas	Area name (set when drawing) must match the location column value exactly.
Incident Count heatmap	Analysis tab → Heatmap Mode → Incident Count | Gray = fewer, Blue = more
Distance/Time heatmap	Requires: station marker + timeElapsed column | Red = slower, Green = faster
Standard filter	Analysis tab → Data Filters → expand a column → check values
Numeric custom filter	Analysis tab → Custom Filters → + Create → 📊 Numeric Ranges → format: <18;18-30;>30
Categorical custom filter	Analysis tab → Custom Filters → + Create → 🏷 Categorical Groups → check values → name group → + Add Group
Group locations	Settings → Categories → Enable Category Mode → Start New Category
Cancel a drawing	Press Escape at any time while drawing is active
Undo a drawing point	Ctrl+Z (or Cmd+Z on Mac) when drawing
Delete an area	Right-click inside the finished polygon
Clear all areas at once	Settings → Area Selection → 🗑 Clear All Areas
Clear all categories at once	Settings → Categories (Category Mode on) → 🗑 Clear All Categories
Back up all work	Settings → Import/Export → Export All Settings → save the .json file